

Data Structures and Algorithms Spring 2024 — Problem Sets

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Week 7. Problem set

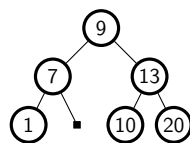
1. Insert the following keys into an initially empty AVL tree [Goodrich, Section 11.3]:

3, 29, 5, 11, 23, 13, 17, 19, 7, 2

For each insertion:

- show the *state of the tree* **after** the insertion
- specify the *number of rotations* performed during the insertion

Depict each tree using the array representation for binary trees. For example, consider the following AVL tree:



The tree above must be depicted as the following array:

9	7	13	1	—	10	20
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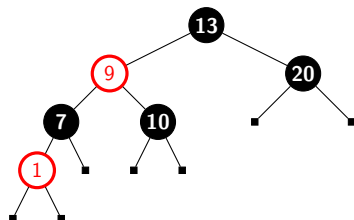
2. Insert the following keys into an initially empty red-black tree [Cormen, Chapter 13]:

3, 29, 5, 11, 23, 13, 17, 19, 7, 2

For each insertion:

- show the *state of the tree* **after** the insertion
- specify the *number of rotations* performed during the insertion

Depict each tree using the array representation for binary trees. You **must** color the keys for red nodes with **red** and keys for black nodes with **black** color. For example, consider this tree:

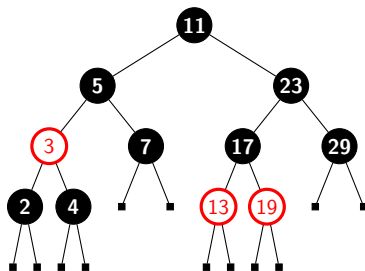


The tree above must be depicted as the following array:

13	9	20	7	10	—	—	1	—	—	—	—	—	—
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3. Delete the following keys (in this order) from a red-black tree [Cormen, Chapter 13] given below:

3, 29, 5, 11, 23



For each deletion:

- show the *state of the tree* **after** the deletion
- specify the *number of rotations* performed during the deletion

Depict each tree using the array representation for binary trees. You **must** color the keys for red nodes with **red** and keys for black nodes with **black** color. For example, the starting tree for this exercise should be depicted as the following array:

11	5	23	3	7	17	29	2	4	–	–	13	19	–	–
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References

- [Cormen] T. H. Cormen, C. E. Leiserson, R. L. Rivest and C. Stein. *Introduction to Algorithms, Fourth Edition*. The MIT Press 2022
- [Goodrich] M. T. Goodrich, R. Tamassia, and M. H. Goldwasser. *Data Structures and Algorithms in Java*. WILEY 2014.